Architesh Prasad

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EDUCATION

California Institute of Technology, Pasadena Ca *Expected 2027*
Chemical Engineering (Computational Track); minor in Computer Science (GPA: 3.9)

PUBLICATIONS

Protein Structure Model Embeddings Encode Interpretable Signals of Multistate Behavior 2026

Architesh Prasad (sole author)
NeurIPS 2026 Workshop on Interpretability for Discovery — Accepted

Multi-omics liquid biopsy identifies mitochondrial dysfunction in geographic atrophy and supports intraocular bioavailability of α-ketoglutarate 2026

Tsai-Chu Yeh, Gabriel Velez, **Architesh Prasad**, …, Vinit Mahajan
Nature Communications Medicine — Accepted

EXPERIENCE

Independent AI for Science Researcher

May 2026-Present

- Conducting independent machine learning research focused on applications in protein modeling and AI for scientific discovery. Projects have examined protein dynamics, generative structure models, and interpretability in models including AlphaFold, ESM, and diffusion-based protein models.
- Developing research pipelines for dataset construction, model evaluation, representation analysis, and ablation studies. Current work focuses on understanding model failure modes and designing computationally efficient methods for evaluating alternate protein conformations.
- Research has led to multiple ongoing manuscripts and an accepted NeurIPS 2026 workshop paper, with projects spanning benchmark development, protein conformational modeling, and analysis of learned representations.

Undergraduate Research Fellow *Palo Alto, Ca & Virtual*
Stanford University *June 2025-Present*

- Leveraging bioinformatics and machine learning for multiomic data in the Mahajan Lab. Assisting with multiple projects to guide future experiments and present insights from the data. Projects involve research targeting diseases including Age Related Macular Degeneration and Retinitis Pigmentosa.
- Initial results have led to early stage writing for three independent manuscripts with focuses on both new bioinformatics applications and clinically relevant results.
- More generally, working on developing bioinformatics pipelines that enable any lab member to import multi modal data and answer more complex data analysis questions such as information on differentially expressed proteins between treatment groups or general trends in protein groups.

John Stauffer SURF Fellow *Pasadena, Ca*
California Institute of Technology *June 2024–August 2024*

- Performed summer research under the mentorship of the Shan Lab at Caltech. Designed a plasmid library and conducted golden gate assembly to supplement the directed evolution of chaperone proteins. Applications in targeting protein aggregates, including those involved with Alzheimer's and Parkinson's disease.
- Plasmids produced by this library via a testing construct showed anti-aggregation activity against alpha-synuclein aggregates, a protein associated with Parkinson's disease.

Fellow *Virtual*
Non-trivial *June 2023–September 2023*

- One of 100 selected from over 20,000 applicants to work on a project concerning existential risks including biosecurity and superintelligent AI. Co-led a literature review and analysis of genetic engineering. Interviewed Dr. George Church of Harvard for recommendations and feedback. Group focused on developing guidelines for the safe emergence of commercial-scale gene editing.

SKILLS

Technical
Languages: Python, MATLAB, R ML/AI: PyTorch, scikit-learn, transformers, diffusion models, CNNs, representation learning **Data/Scientific Computing**: NumPy, pandas, statistical modeling, PDE modeling, optimization **Bioinformatics**: protein design, multiomic analysis, sequence analysis **Tools**: Git, LaTeX, ChemDraw

Laboratory
Spectroscopy, PCR, Golden Gate Assembly, Gibson Cloning, Protein Purification, Gel Electrophoresis, Dilution Spot Plating, Chemical Transformation, Electroporation

RELEVANT COURSEWORK

CS148: Deep Learning and LLMs
CS155: Machine Learning & Data Mining
CS156: Learning Systems
ChE137: Data Science for Chemical Systems
ChE15: Intro to Chemical Engineering Computation
ChE63: Thermodynamics
ChE103: Heat, Momentum, & Mass Transport
Ch41: Organic Chemistry
BEM103: Introduction to Finance
Bi8: Molecular Biology
ACM95: Complex Analysis & PDEs
ACM11: Intro to Computational Science
ACM104: Applied Linear Algebra
Ma2: Differential Equations
Ma3: Probability & Statistics
IDS157: Statistical Inference
BE163: Intro to Biomolecular Engineering
Ec122: Econometrics

OTHER EXPERIENCES

Ch1b TA *Pasadena, Ca*
California Institute of Technology *January 2025–April 2025*

Teaching assistant for Chem 1b, Caltech's introductory chemistry course. Working on course development by the production of demonstration videos, live experiments, and writing problem set questions. Class consists of over 200 students.

Social Media Ambassador for Caltech Admissions *Pasadena, Ca*
California Institute of Technology *May 2024–Present*

Photographer and editor for the undergraduate admissions office at Caltech. Produced videos and photographed events for platforms including Instagram and the Caltech Admissions website.

SELECTED PROJECTS

LSER Model for Blood to Tissue Partitioning 2025
Built a model using linear energy solvation relationships (LSER) to predict the partitioning of molecular pharmaceuticals from blood to general tissues. Consists of an entire data science pipeline starting from data cleaning and ending with the construction of a prediction model for LSER characteristics based on thermodynamical properties. Project was conducted through ChE137: Data Science for Chemical Systems.

Implementing Machine Learning methods for Proteomic Data 2025
Proteomic data is challenging in that the number of features often vastly exceeds the number of labels. Since reliable generation of these datasets has been a new paradigm, machine learning questions relating to these datasets have not been persued. This project aims to test & adapt many different ML approaches onto proteomic data to produce a framework for building models. Feature selection, neural architectures, bootstrapping, & adaptations on reccomender models are among the methods tested. These models could be deployed for use cases including biomarker identification & diagnosis.